

Lab Task-1: Solving Puzzles using exhaustive search**(10 marks)**

Brief introduction to the problem of 15-puzzle. Representation of the problem states and implementation of solution using DFID (depth first with iterative deepening).

Lab Task-2: Solving Puzzles using informed search**(10 marks)**

Improve the exhaustive search solution in task-1 using problem heuristics. Implement various (at least 3) heuristic functions with A* algorithm to make the solution efficient. Compare the performance of exhaustive versus heuristic methods.